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Setting up the PM/RM/CM Controller for A Generic PM Motor

Revision 2.7

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1. Overview

This document describes how to configure the PM/RM/CM Controller Software to operate a generic surface magnet permanent magnet motor (PM). It also discusses how direct Id/Iq control can be used for testing of IPM motors.

The PM Family of inverters comes in three variants, Gen 2, Gen 3, and Gen 5. Gen 2 and Gen 3 have nearly the same functionality, Gen 2 hasn't been produced since roughly 2010. Most PM Family inverters produced are Gen 3. The Gen 5 uses a different processor target and has a different resolver interface than the Gen 3.

The RM family (Gen 4) is code compatible with PM Family Gen 3 utilizing the same processor target.

The CM family is code compatible with Gen 5, they share the same processor target. The CM family uses a different resolver interface than the RM family and Gen 5 family.

The PM/RM/CM software implements a generic indirect field-oriented control algorithm (IFOC).

A full description of IFOC is beyond the scope of this manual. For IFOC to work properly it is important to have accurate motor parameters. These parameters are used to calculate the correct motor voltage to operate the motor.

****** Important Note:** Because the motor type is set to 65 the firmware used must be Group 2 firmware. Implementation of this feature occurred in software

version 1942. For RM inverters the implementation was begun with all versions starting from 2000.

All Gen 5 firmware (version 65xx) supports this motor type 65.

****** Important Note:** The RM product line requires firmware versions that begin with 2XXX. The RM product line will **not** work with 19XX firmware. In most respects the RM firmware behaves identical to the normal PM 19XX firmware.

This software will not work to properly run an Interior Permanent Magnet Motors (IPM Motor) on a vehicle. The software will not automatically provide the angle advance that an IPM needs to utilize the reluctance torque. This software can be used to test an IPM motor on a dynamometer by using the direct id and iq commands. Full support of an IPM motor is done through custom id/iq tables that are built into custom firmware. Our existing firmware does support many IPM motors, e.g., BorgWarner HVH250 family.

General Setup Notes:

- Gather necessary motor parameters
- Open the special defsyms file “defsyms_PM_Motor.txt” using the RMS GUI software. The defsyms file is opened by using the File menu that is in the upper left corner of the Memory Window of the RMS GUI. This allows access to special EEPROM parameters. Note that the defsyms_PM_motor.txt file will not be the same between Gen 3 and Gen5/CM inverters.
- Program EEPROM parameters, including setting the motor type to 65
- Verify that the resolver feedback is working correctly by monitoring the speed feedback.
- Perform the resolver calibration
- Test the motor to ensure desired performance

2. Setting the ‘Motor Type’

The PM/RM/CM Family firmware has the ability to drive many different motor types. The user must tell the controller what type of motor it is to be used with. This is done via the Motor_Type_EEPROM parameter. Refer to the Programming EEPROM Parameters using GUI manual for more information on programming EEPROM parameters.

Motor Info	Motor Type
Generic PM Motor	65

When you set the motor type via the GUI it will automatically adjust some default parameters. However, there are several additional parameters that should be reviewed and adjusted if desired.

EEPROM Parameter	Description
Veh_Flux_EEPROM_(Wb)_x_1000 CAN Parameter Addr: 106	This parameter is used to set the back EMF constant for the motor. The back EMF constant is used for calculating the expected motor voltage as well as the amount of torque generated for a given amount of current. In a PM motor Torque = 3 * Poles / 4 * flux * iq. Flux is in terms of Vpk/rad/s, iq current is interms of peak amps. The flux is set as a x1000 value, so for example a flux of 0.102 would be set as a value of 102.
Angle_Advance_Factor_EEPROM_x100 CAN Parameter Addr: 176	This parameter provides an angle advance that increases with speed. In some cases, providing the angle advance helps align the D and Q axis so that it matches the internal motor model. The setting is scaled in time, where a setting of 100 would be a value of 1 PWM period. A setting of 50 would be ½ PWM period. See section 2.11 for more details.
Gamma_Adjust_EEPROM_(Deg)_x_10 CAN Parameter Addr: 152	Used with the resolver angle offset calibration process. See the Resolver Calibration Manual. The resolver angle offset must be calibrated before operating the motor at anything but a greatly reduced torque command.
Resolver_PWM_Delay_EEPROM_(Counts) CAN Parameter Addr: 151	Used with the resolver angle offset calibration process. See the Resolver Calibration Manual.
IQ_Limit_EEPROM_(Amps)_x_10 CAN Parameter Addr: 100	This parameter sets the maximum value of the torque producing current that can be commanded. The amount of allowable current is dependent on the motor type and the controller type. The controller is the limiting factor for the motor current. Set the IQ_limit to be 636 Apk (450Arms). The current is set in peak Amps times 10. For example, 300 A_{rms} = 4250.
ID_Limit_EEPROM_(Amps)_x_10 CAN Parameter Addr: 101	This parameter sets the maximum ID current. In a PM motor this ID current is the field weakening current. Generally when beginning to test a PM motor it would be best to start at a very low value of ID current, or zero.
Mtr_OverTemp_Limit_EEPROM_(C)_x_10 CAN Parameter Addr: 113	This parameter sets the motor over-temperature fault limit. It is set in degrees C times 10. Thus for 150°C it would be set to 1500.

Motor_Overspeed_EEPROM_(RPM) CAN Parameter Addr: 111	This parameter sets the speed at which an over-speed fault will be generated. It should be set based on the needs of the vehicle system for over-speed protection.
Max_Speed_EEPROM_(RPM) CAN Parameter Addr: 128	This parameter sets the maximum speed that the controller will command.
Break_Speed_EEPROM_(RPM) CAN Parameter Addr: 127	The Break Speed is the speed at which the torque capability of the motor starts to decrease due to lack of voltage from the motor controller. This value should be set based on information from either Cascadia Motion or the motor manufacture on the performance of the motor with the particular battery voltage being used.
Motor_Torque_Limit_EEPROM_(Nm)_x_10 CAN Parameter Addr: 129	This sets the maximum torque command.
Regen_Torque_Limit_EEPROM_(Nm)_x_10 CAN Parameter Addr: 130	This sets the maximum regen torque command.

To setup the specific motor the following parameters must be set

EEPROM Parameter	Description
Number_of_Poles_EEPROM CAN Parameter Addr: 205	This parameter sets the number of motor poles. It is used for properly calculating the motor speed as well as for interpreting the resolver. Must be an even number (2,4,6,8, etc.).
Number_of_Resolver_Poles_EEPROM CAN Parameter Addr: 206	This is the number of poles of the resolver. Resolver manufacturers typically use a notation like 1X, 2X, 3X, etc. This corresponds to 2 poles, 4 poles, 6 poles, etc. The number of motor poles divided by resolver poles must be an integer. For example 10 motor poles and 4 resolver poles is not allowed. Generally the best performance is when the number of resolver poles is equal to the motor poles.
Resolver_Exc_Level_EEPROM CAN Parameter Addr: 207	Sets the Excitation level of the resolver. **PM Family Inverters:** The range of values is from 3 to 7. 3 represents the largest signal and 7 represents the smallest. The resolver excitation level should be set so that the sin/cos feedback is no larger than about 1.2Vpk **RM Family Inverters:** The RM Family inverter only has two levels of excitation. The high level is set by an excitation value of 3. The low level is set by an excitation value of 5. **CM Family Inverters:** The CM family inverter has more options for the excitation level. See further details below.
Natural_Frequency_EEPROM_(rad/sx10) CAN Parameter Addr: 208	The Natural Frequency is the frequency response of the observer (PLL) used with the resolver. The observer filters noise from the resolver. A typical value would be 500 rad/s (setting of 5000). The typical range of values would be from 100 rad/s to 1000 rad/s.
Damping_Factor_EEPROM_x_100 CAN Parameter Addr: 209	This is the damping factor used with the resolver observer. A typical value would be 1.1 (setting of 110). The range of values are typically around 0.8 to 1.2.
Resolver_Fault_Count_EEPROM_(0=disable) CAN Parameter Addr: 231	The resolver fault is normally presented only after several incorrect readings. This parameter controls how many incorrect readings are needed to generate a fault. The normal default value is 5. If it is desired to disable this fault then set this value to 0.
V_Base_EEPROM_(Volts) CAN Parameter Addr: n/a	This parameter defines the voltage base used for internal per unit calculations. Generally this parameter is set to 200 representing 200V.

I_Base_EEPROM_(Amps) CAN Parameter Addr: n/a	This parameter defines the current base used for internal per unit calculations. Generally this parameter is set to 300 representing 300A.
F_Base_EEPROM_(Hz) CAN Parameter Addr: n/a	This parameter defines the frequency based used for internal per unit calculations. Generally this parameter is set to 175 representing 175Hz.
Motor_Type_Options_EEPROM CAN Parameter Addr: 175	This parameter represents a 16-bit binary number. Bit 0: If set to 1 the torque will be reduced by 1/speed^2 when above Break Speed, if set to 0 then the torque will be reduced by 1/speed. Bit 1: If set to 1 the current regulator cross coupling is active, if set to 0 is inactive. Normally this is set to 1. Bit 2: If set to 1 software will allow resolver offset calibration using the Sin_Offset_EEPROM_(ADC_Counts) and Cos_Offset_EEPROM_(ADC_Counts) parameters. Bit 3: Alternative field weakening controller. Must be set to 0 except when using Direct Id and Iq commands (bit 10). Changes Modulation index calculation by sqrt(3)/2 (maximum = 1.0) Bit 4: If set to 1 the Sin and Cos inputs of the resolver will be flipped in software. This will result in the rotation direction for positive speed to be flipped. Bit 5: If set to 1 enables field weakening controller. Normal value for this is to set to 1. This bit must be set to 0 when the alternative field weakening bit (bit 3) is set to 1. Bit 6: If the motor is a high-speed motor, generally faster than about 30krpm, then set this bit to 1. All speed feedback variables will be changed from rpm to rpm /10. Bit 7: If this bit is set to 1 it enables AIN3 (normally the brake) to be used to adjust an angle offset that is added to the gamma_adjust parameter. This angle offset can be used for testing of IPM style motors. Bit 8: Reserved, set to 0 Bit 9: Reserved, set to 0 Bit 10: Enable Direct Id and Iq commands from CAN. Bit 11: RM, CM, and Gen5 inverters use a resolver-to-digital converter IC for resolver. To enable use of this IC this bit must be set to 1. For PM inverters this bit must be set to 0. For special versions of RM, CM, and Gen5 inverters, this bit can be set to 0 to use a sin/cos encoder. Bit 12: To enable Saturation Effect set to 1. Bit 13: To enable Phase Advance set to 1. Bit 14: To enable sin/cos encoder offset features set this bit to 1. Bit 15: Allows swapping of Phase A and B in firmware if set to 0. This will reverse the direction (sign) of the Voltage Feedback Speed. This can be helpful if it is desired to change the direction of rotation that causes a positive Voltage Feedback Speed.

Examples of the setting of the Motor Type Options EEPROM Parameter: Typical value for this parameter for a PM inverter would be 0b0000 0000 0010 0010, or 34 in decimal. Typical value for this parameter for a RM inverter would be 0b0000 1000 0010 0010, or 2082 decimal	
Ld_Lq_Const_EEPROM_(uH) CAN Parameter Addr: 210	This is the d-axis inductance of the motor (Ld). It is set in terms of microhenries. For example 4.9mH would be set as 4900. The same value is used for Lq as well.
Stator_Resistance_EEPROM_(Ohmsx10000) CAN Parameter Addr: 211	This is the per phase stator resistance. Resistance is set as ohms times 10000. So a resistance of 0.0343 ohms is set as 343.
Maximum_Torque_EEPROM_(Nmx10) CAN Parameter Addr: 212	This parameter is used to set the maximum torque commands that are allowed to be used with this motor.
Mod_Limit_EEPROM_x_100 CAN Parameter Addr: 213	This parameter defines the modulation limit reference used with the field weakening controller. If the modulation level exceeds this value then the controller will begin added Id current to field weaken the motor. If the current requirement exceeds the Id current limit then the Iq current will be reduced. If the Iq current then reaches zero then the controller has no options for reducing the motor voltage further. Operating at speeds above this level will likely cause a loss of control of the motor. Generally this parameter is set to a value of 1.10, giving about 5% margin to the maximum value (1.15). The range of values is 0.9 to 1.15. The parameter is set in terms of the modulation index, 115 presents 100% modulation. If Direct Id/Iq current mode is being used, then the modulation index scaling is changed so that 100% modulation is a value of 1.0. In this case a typical modulation limit would be 0.95 (a setting of 95)
Mod_Hi_Max_EEPROM_x_100 CAN Parameter Addr: n/a	Above this modulation level the current regulator has no effect. The voltage is being commanded based on feedforward equations. This value should be set to 140.
Mod_Lo_Max_EEPROM_x_100 CAN Parameter Addr: n/a	Below this modulation level the current regulators are in full control of the motor voltage. Between the level and Mod_Hi there is a phase out of the current regulator. This value should be set to 120.
Kp_Current_EEPROM_x_1000 CAN Parameter Addr: 214	This parameter is the Proportional Gain of the Current Regulator. A gain of 0.2 is a setting of 200.

Gen 3: Ki_Current_EEPROM_x_10000 Gen 5/CM: w1_Current_EEPROM_(rad/s) CAN Parameter Addr: 215	This parameter is the Integral Gain of the Current Regulator. A gain of 0.02 is a setting of 200. The bandwidth of the current regulator (w1) = Ki / (Kp * T). Where T = 1/Fpwm (12kHz). For example if Ki = 0.02 and Kp = 0.2 then bandwidth = 1200 rad/s (191 Hz) For Gen 5/CM inverters the bandwidth is directly entered into the EEPROM parameter and the Ki value is automatically recalculated for the PWM frequency that the inverter is operating at. So in this example instead of entering a value of 20 (representing Ki = 0.02) use a value of 1200 for w1.
Kd_Current_EEPROM_x_1000 CAN Parameter Addr: 216	This parameter is the Derivative Gain of the Current Regulator. A gain of 0.2 is a setting of 200. Normally this is set to 0.
Klp_Current_EEPROM_x_1000 CAN Parameter Addr: 217	This parameter sets the frequency bandwidth of a low pass filter applied to the PID regulator. The filter frequency is w2 = Klp / T, where T = 1/Fpwm (12kHz typical). Normally this value is set to 0.
Kp_Current_Slope_EEPROM_x_100 CAN Parameter Addr: 218	The Kp Current Slope and Offset allow the proportional gain of the current regulator to be modified with speed. This feature is not normally needed. These two parameters should be set to a value of 0.
Kp_Current_Offset_EEPROM_x_1000 CAN Parameter Addr: 219	See above.
Kp_Flux_Weakening_EEPROM_x_1000 CAN Parameter Addr: 220	This parameter is the Proportional Gain of the Field weakening Regulator. A gain of 0.8 is a setting of 800. The typical default gain for this parameter is 0.8.
Gen 3: Ki_Flux_Weakening_EEPROM_x_10000 Gen 5/CM: w1_Flux_Weakening_EEPROM_(rad/s) CAN Parameter Addr: 221	This parameter is the Integral Gain of the Field Weakening Regulator. A gain of 0.02 is a setting of 200. The bandwidth of the regulator (w1) = Ki / (Kp * T). Where T = 1/Fpwm (12kHz typical). For example if Ki = 0.02 and Kp = 0.2 then bandwidth = 1200 rad/s (191 Hz). The typical default gain for this parameter is 0.002. For Gen 5/CM inverters the bandwidth is directly entered into the EEPROM parameter and the Ki value is automatically recalculated for the PWM frequency that the inverter is operating at. In this example instead of entering a value of 20 (representing Ki = 0.02) use a value of 1200 for w1.
Kd_Flux_Weakening_EEPROM_x_1000 CAN Parameter Addr: 222	This parameter is the Derivative Gain of the Field Weakening Regulator. A gain of 0.2 is a setting of 200. Normally this is set to 0.

Klp_Flux_Weakening_EEPROM_x_1000 CAN Parameter Addr: 223	This parameter sets the frequency bandwidth of a low pass filter applied to the Field Weakening regulator. The filter frequency is w2 = Klp / T, where T = 1/Fpwm (12kHz typical). Normally this value is set to 0.
Motor_Temperature_Sensor_Type_EEPROM CAN Parameter Addr: 224	This parameter allows selection of the motor temperature sensor type. By selecting this it will configure which input is used and how it is converted to temperature. All sensors are connected to AIN2 unless otherwise defined. Currently defined types: 0: Siemens motor used on Ford Ranger 1: 30K NTC Thermistor w/ 1K pull-up (Remy) 2: KTY 84/130 PTC w/ 1K pull-up 3: Report Motor Temperature as 0 always 4: The higher of RTD1 and RTD2 (PT1000) 5: 49.12K @ 25°C NTC Thermistor w/ 1K pull-up 6: NTC used on MES motor 7: KTY 81-210 PTC w/ 3K pull-up 8: 10K NTC Thermistor with 1K pull-up 9: 100K NTC Thermistor 10: Reserved 11: 33K NTC Thermistor w/ 4.7K pull-up 12: Use RTD1. 13: 1K @ 100C NTC Thermistor w/ 1K pull-up 14: 220K @ 25C NTC Thermistor w/ 10K pull-up 17: 2x 33K OHM NTC connected to AIN2 and AIN4 w/ 4.7K pull-up 24: 176 ohm @ 180°C NTC thermistor connected to AIN2 with 1K pull-up resistor 25: 10K @ 25°C NTC Thermistor B25/85=3986K connected to AIN2 with 1K pull-up resistor. 27: Two sensors, same type as sensor type 25, connected to AIN2 and AIN4. Motor temperature is the highest of the two temperatures. Version 2028. 28: Two sensors, same type as sensor type 25, connected to AIN2 and AIN4. Motor temperature is the highest of the two temperatures. Version 2028. 32: iM-225/SS-250 two embedded temperature sensors with oil temperature and pressure reporting. 35: iM-225/SS-250 single embedded sensor with oil temperature and pressure reporting. 36: iM-225/SS-250 two ring terminal temperature sensors with oil temperature and pressure reporting. 38: iM-425 embedded temperature sensor with oil temperature and pressure reporting.
Saturation_Gain_EEPROM_x_10 CAN Parameter Addr: 228	This is the Saturation Gain parameter. Used to provide gain in current command compared to the theoretical value. A gain of 1.12 would be set with a value of 112.
Saturation_Torque_EEPROM_(Nm)_x_10 CAN Parameter Addr: 229	The Saturation Slope parameter is applied for torque commands above the Saturation Torque. Value is set in Nm times 10, so 120Nm would be set as 1200.

Saturation_Slope_EEPROM_(1/Nm)_x_10000 CAN Parameter Addr: 230	The Saturation slope define rate of increase in gain above the Saturation Torque. For example to apply a gain of 0.0040 then the setting would 40.
Phase_Advance_EEPROM_(deg/Nmx1000) CAN Parameter Addr: 232	The Phase Advance value controls the amount of phase advance that occurs with torque command. The value is set in terms of degrees per Nm times 1000
Angle_Advance_Factor_EEPROM_x_100 CAN Parameter Addr: 176	This is a calibration parameter used in the alignment of the magnetic field of the motor by adjusting the angle based on speed. The setting is scaled in time, where a setting of 100 would be a value of 1 Control period. A setting of 50 would be ½ Control period. See Section 2.11 for more details.

2.1 Settings for Parameters

Internally all motor calculations are done with a per unit system. A typical per unit base would be a Current Base of 300A. Internally then 300A would have a value of 1.0. From most all motors that would operate with the PM motor controllers using per unit bases of 300A, 200V, and 175Hz is appropriate. In some cases if the motor torque is quite high it may make sense to adjust the per unit bases. Consult with Cascadia Motion for more information.

The settings of the parameters depends on knowledge of the motor parameters.

The stator resistance is the phase to neutral resistance of the motor. This can be found by measuring the line-to-line resistance and dividing it by 2.

The Ld and Lq values are assumed to be the same. These values typically come from the motor manufacturer.

The flux value can be calculated from the back EMF constant of the motor. The flux value is set in terms of Vpk / rad/s. The Vpk value is the voltage from line to neutral in terms of peak volts (RMS times the sqrt(2)). The rad/s value is the electrical angle, not the mechanical angle. If the mechanical angle is known, then the electrical angle can be found by multiplying the poles/2.

Here is an example of setting the flux value. Assume a 12 pole motor that has a voltage constant of 48.39 Vrms/krpm. The Vrms is in terms of Vrms line to line. To convert the voltage to Vpk line to neutral we multiply by sqrt(2) and divide by sqrt(3), so 48.39 * sqrt(2) / sqrt(3) = 39.51Vpk l-n. 1krpm = 1000 rpm. The electrical frequency would be 1000 * 12 / 120 = 100Hz, we multiply by 2*PI to get 628.3 rad/s. So the resulting flux value would be 39.51 / 628.3 = 0.063.

The modulation limiter used in the software can be a bit confusing. The modulation index is a measure of how much voltage the controller is producing. A value of 1.15 is the maximum voltage that can be produced without going into over-modulation (for IPM motors the maximum is shifted to 1.0). Above 1.15 the control of voltage becomes non-linear. Thus to keep the current regulators working in a linear area the normal maximum modulation limit is set to 1.10 (in some cases it could be set lower than this).

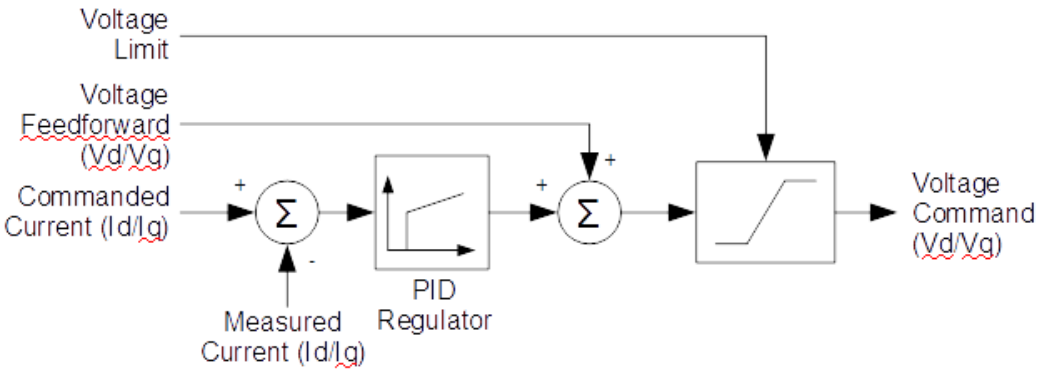
The modulation limiter acts to add negative Id current. That is as the motor speed or torque is increased if the modulation limit is reached then the control automatically adds negative Id current. If the Id current limit is reached (EEPROM parameter) then the controller must reduce the Iq current command in an attempt to keep the modulation level at the modulation limit. If the Iq current reaches zero, then the system has exceeded the ability of the control to keep the

modulation level under control. This situation should be avoided. That is the system should not be designed such that the Id limit is reached at an Iq current of zero.

In a steady state sense, there is a theoretical limit to the Id current. This theoretical value is calculated as Flux_pm / Ld (in peak amps). In theory at this point additional Id current would actually increase the voltage and thus lead to an unbounded (or stuck) modulation limiter. In practice the maximum value of Id current tends to be substantially less than theoretical value. Generally, an Id limit of about 50% of the theoretical limit works well.

2.2 Current Regulator

The motor control software uses two current regulators. One regulator for the Id current and one regulator for the Iq current. Both regulators have the same structure and use the same regulator gains.



The current regulator algorithm is executed at the PWM frequency. For most PM Family software, the PWM frequency is 12kHz. Thus the current regulator is executed every 83.33 microseconds.

The PID inputs and outputs are done using per-unit calculations. If needed contact Cascadia Motion for the base units for the particular motor being operated.

The most common set of per unit bases are as follows:

Voltage Base = 200V
Current Base = 300A

So when attempting simulate or model the PID regulator the inputs and outputs must be first scaled by the per unit bases. For example, if the Commanded

Current was 150A then the per unit commanded current would be 150 / 300 = 0.5. The output voltage from the PID regulator would have to be multiplied by the Voltage Base to get the actual voltage command. For example if the voltage command was 1.0 in per unit the actual voltage commanded would be 1.0 * 200 = 200V.

A starting point for the current regulator gains can be found by using the following analysis.

Assume a controller bandwidth w1. A value for w1 is round as Rs / Ldq. Rs is the stator resistance in ohms. Ldq is the Ld = Lq value.

A value for Kp (proportional gain) can then be calculated as Ldq * w1 / (Vbase/Ibase). Again for a typical setup of Vbase = 200 and Ibase = 300, then Vbase/Ibase is 0.6667.

The Ki value (integral gain) can the be calculated as Ki = Kp * w1 * Tpwm. The current regulator is operated at the PWM frequency of 12kHz, so Tpwm = 1/12kHz.

On Gen 5 and CM inverters the Ki value is calculated from the w1 EEPROM parameter because the Tpwm can change as the PWM frequency changes.

2.3 Counteracting Saturation Effect (applies to firmware after July 11, 2018)

For many motors there exists a motor saturation effect. The actual torque produced by the motor does not follow the classic motor equation when the current exceeds a certain value.

The linear equation for motor torque is:

Torque (Nm) = 3 * Poles / 4 * Iq * $Flux_{PM}$

To provide more accuracy between the commanded torque and the actual torque produced it is necessary to account for the fact that the FluxPM value is not actually a constant. Additionally there are losses in the motor that can cause the actual torque measured to be less than the theoretical value predicted by the linear torque equation.

If it is desired to use this saturation effect model then first it must be enabled by setting bit 12 of the motor type option EEPROM parameter to 1. If the bit is set to 0 then the saturation effect model will not be applied.

The control software provides three parameters for addressing this:

- Saturation Gain (Gsat)
- Saturation Torque (Tsat)
- Saturation Slope (Msat)

These three parameters are used to modify the $Flux_{PM}$ of the torque equation. The modification takes the form:

$$Flux_{PM} = Flux_{PMnom} * f(Tcmd)$$

FluxPMnom is the nominal PM flux for the motor. Tcmd is the torque command.

So f(Tmd) is now a gain factor against the nominal flux. And typically this would have a setup that decreases the flux above a certain torque.

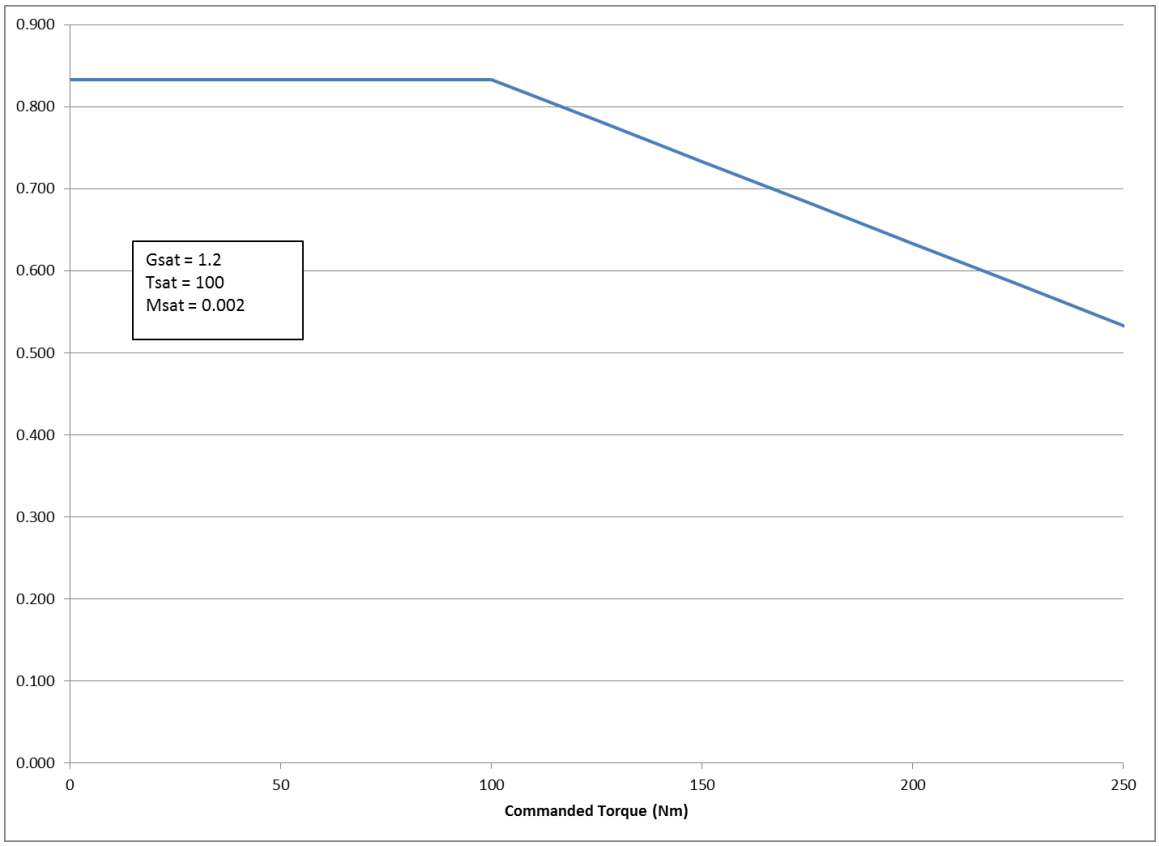
The new flux thus follows the following formula:

f(Tcmd) = Msat(Tsat-abs(Tcmd)) + 1/Gsat for abs(Tcmd) > Tsat
f(Tcmd) = 1/Gsat for abs(Tcmd) < Tsat

The Saturation Gain (Gsat) provides a multiplication factor on the conversion of a torque command to a current command. If the Saturation Gain is 1.00 then it would provide no multiplication. If the Saturation Gain is 1.20 then the current command will be 1.2 times higher than the theoretical model would predict. The Saturation Gain modifies the Flux by a 1/Gsat effect. So if Gsat is greater than 1 then it is lowering the flux (effectively lowering the torque constant of the motor).

The Saturation Torque defines the amount of torque command that exists before the Saturation Slope is applied. Below the Saturation Torque the only effect is the Saturation Gain parameter, above the Saturation Torque the Saturation Slope is also applied.

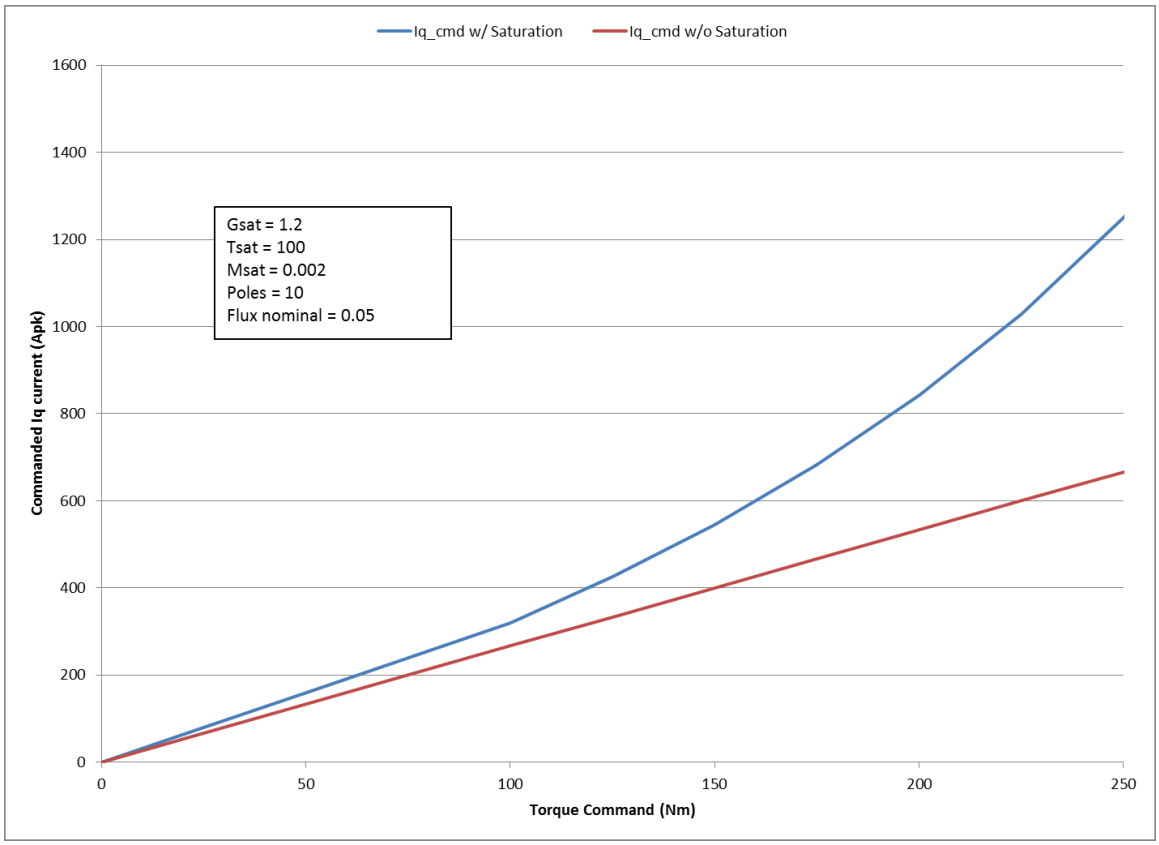
The Saturation Slope (Msat) defines an additional gain that is applied that decreases the flux value with increases in Torque Command above the Saturation Torque (Tsat). The Saturation Slope is defined in terms of 1/Nm (per Nm). So for example a Saturation Slope of 0.004 increases the overall gain by 0.004 per Nm.



Example graph of the f(Tcmd) given the example parameters.

The above graph shows an example of how the additional gain would be applied given some example parameters.

Below is a graph that shows the effect that this gain factor would have on the commanded motor current



Example of the Saturation Effect on the commanded current.

If it is desired that this effect not be used then the Saturation Gain should be set to 1.0 and the Saturation Slope should be set to 0.0.

2.4 Phase Advance – implemented in 19B8 firmware

Phase Advance refers to a shift in the D-Q vector axis relative to the position information given by the resolver.

The Angle Advance Factor provides this phase shift in terms of speed. That is for an increasing speed the offset angle is increased linearly. Normally this is used to account for phase shift effects related to the resolver or related to the control as the fundamental frequency of the motor increases. The Angle Advance Factor is always used regardless of whether the Bit 13 of the Motor Type options is set or not.

In some cases there are motors that are not controlled as an IPM motor would normally be controlled but do need to have a certain phase shift to optimize the amount of torque produced. This type of control would often be considered more of a Phase Advance type control. That is the motor control angle is increases in proportion to the Torque command.

To enable Phase Advance set Bit 13 of Motor Type Options EEPROM parameter to 1.

Then the parameter Phase_Advance_EEPROM sets the amount of phase advance that occurs. The Phase_Advance_EEPROM parameter is set in terms of degrees per Nm of torque times 1000.

When the phase advance is operating it results in a shift of the Iq and Id current commands. Internally the output of the torque regulator generates a total current command. The total current command will be proportioned between the Iq and Id currents as:

Iq_cmd = Total_current_command * cos(phase_advance_angle)
Id_cmd = abs(Total_current) * sin(phase_advance_angle)

The phase_advance_angle is calculated as:

phase_advance_angle =
abs(torque_command)*Phase_Advance_EEPROM/1000

The phase_advance_angle is always between 0 and -90 degrees. The amount of phase advance can be monitored from the RMS GUI with the defsyms_PM_motor file using the Memory Window parameter Phase_advance.

The total current command is calculated from the following equation:

Total_Current_Command = Torque Command / (3*P/4 * flux_cmd)

The torque feedback calculation is modified when using the Phase Advance feature. The new calculation is:

Torque = sign(iq feedback) *3*P/4 * flux_cmd * mag(Iq feedback, Id feedback)

So the revised torque equation does is not strictly correct in the sense that it does not use differences in Ld/Lq to calculate the reluctance torque, however, it does take into account the total motor current and not just the Iq portion of the current. Thus this calculation matches the equation that is used to calculate the total current command as above.

2.5 Direct Control of Id and Iq Commands

Direct control of the Id and Iq current commands can be useful when testing an IPM style PM motor.

NOTE: This feature was implemented if firmware version 19A4, it is available in all Gen 5 firmware.

In an IPM style motor it is necessary to provide the optimal Id and Iq current commands for the particular operating point.

There are two regions of operation for an IPM motor. The MTPA (Maximum Torque per Amp) region is where the Id and Iq currents are optimized to produce the maximum torque for a given total motor current (sqrt(Id^2 + Iq^2). The MTPV (Maximum Torque per Voltage) region is where the motor is at a higher speed and the voltage output of the inverter is at the maximum. The Id and Iq values are adjusted to maintain the maximum voltage but still achieve the desired torque.

In MTPA testing a typical method would be to use CAN control to keep the total current constant (sqrt(Id^2 + Iq^2) while adjusting the angle of the current. The test would monitor the torque produced by the motor and search for the maximum torque produced.

In MTPV testing a typical method would be to set a value of Id current and then increase the Iq current until the maximum motor voltage (modulation index target) is met.

The data taken from MTPA and MTPV testing can then be compiled to form an Id/Iq look up table. It is possible to implement the table into the custom firmware and run the IPM motor.

If using this method first enable it via the Motor type options as followings:

Set bit 10 of the motor type options to 1. This setting will allow direct control of the Id and Iq current commands from the CAN message.

When using this control the alternative field weakening regulator bit must also be set to 1 (bit 3) and the normal field weakening regulator (bit 5) must be 0.

When this mode is enabled the Id and Iq commands can be sent directly from the CAN message to the current regulators.

The direction bit will have the effect of flipping the sign of Iq.

If the modulation level (motor voltage) increases to the modulation limit then the controller will attempt to reduce the motor voltage by decreasing the Iq current command. It will not change the Id current command. This is opposite type of field weakening as compared to a non-IPM motor.

When operating an IPM motor above base speed this mode can be valuable as the id level can be fixed and the maximum Iq level for that operating condition can be easily explored.

Note that when in this mode the modulation level is defined a bit differently than when in the normal PM motor mode. The maximum modulation level is 1.0.

Since this mode is directly accessing the Id and Iq commands it is necessary for the sender of the CAN messages to manage the rate of change. That is the CAN sender should not make large changes in Id and Iq commands.

The current commands replace the torque and speed commands in the CAN Command message. Refer to the CAN database file.

Bytes 0 and 1: D-axis Current Command in peak Amps (should always be negative).

Bytes 2 and 3: Q-axis Current Command in peak Amps (positive current refers to motoring and negative to regeneration).

2.6 Control Angle Adjustment

An alternative to the direct Id and Iq command of current is to control the Control Angle of the current. This method is applicable to systems where CAN control is not possible.

The motor type 65 feature is often used for developing the required id/iq table for IPM style motors.

An IPM style motor develops reluctance torque that is proportional to Id * Iq in addition to the normal flux * Iq torque. Thus, it is necessary to find the correct control angle that delivers the correct amount of Id and Iq for any given operating point.

By setting bit 7 to 1 it enables direct control of the angle offset from Analog Input 3 (Brake analog input). This method should not be used when analog braking mode.

The angle offset will be about 0 degrees with an analog input voltage of 2.5V. The angle will be negative (which should basically always be) when the input voltage is less than 2.5V. Typically an input voltage > 2.5V is not used.

The GUI or CAN will display then a corrected Id and Iq value based on the offset angle.

The offset angle can be viewed directly in the GUI command window with the parameter: Theta_Cmd_Offset_(Deg)_x_10.

2.7 Resolver Interface – PM Family

The PM Family resolver interface is different than the RM Family.

The resolver excitation output is a sine wave signal at the PWM frequency (normally 12kHz). The output of the resolver is fed back into the inverter at the SIN and COS differential inputs. The output of the resolver is principally defined by the transformation ratio of the resolver.

The PM Family processor does an amplitude demodulation of the feedback via an analog to digital converter. The amplitude is then available at the sin_corr and cos_corr symbols in the RMS GUI. The amplitude as measured at sin_corr and cos_corr should not be larger than 1.2Vpk (value of +/- 120).

The Excitation voltage level of the PM Family can be changed by the Resolver_Exc_Level_EEPROM parameter. The valid range is shown in the table below. The Excitation should be adjusted to provide the maximum signal to the feedback that does not exceed the required maximum values.

	PWM Frequency in kHz						
EXC LEVEL	6	8	10	12	14	15	16
0	X	X	X	X	X	X	X
1	X	X	X	X	2.82	2.52	2.36
2	X	X	3.41	2.5	1.88	1.66	1.52
3	X	X	3.08	2.21	1.64	1.44	1.28
4	X	3.64	2.4	1.69	1.24	1.08	0.95
5	X	3.36	2.2	1.52	1.13	0.98	0.89
6	X	2.75	1.77	1.22	0.9	0.78	0.72
7	X	2.56	1.66	1.14	0.84	0.73	0.66

The above chart represents the measured excitation level (Vrms) on a Gen 3 unit. In the cells where there is “X” it was found that the excitation is too distorted to be useful.

When changing the excitation on a Gen3 inverter it is important to make sure that the SIN and COS levels are within the limits of ± 1.5Vpk. It is suggested that they should not have a peak value above ± 1.2Vpk.

2.8 Resolver Interface – RM and Gen 5 Family

The RM and Gen 5 families resolver interface is different than the PM Family.

The RM and Gen 5 Families uses a resolver to digital converter. The resolver to digital converter takes care of producing the excitation as well as demodulating the feedback.

Important Note:

Important Note: Because the RM and Gen 5 Family does not look at the SIN and COS values directly the sin_corr and cos_corr values will be at zero and do not have any meaning.

Important Note: The RDC chip used has a maximum tracking frequency of 976.6Hz. The resolver used must be selected so that the maximum speed is below this limit. For example, if a 4X (8 pole) resolver is used the maximum speed is 14648 rpm. If the resolver frequency is a limiting factor then consider using a resolver with less poles. Keep in mind that motor poles divided by resolver poles must be an integer.

The resolver to digital converter provides feedback to the processor if there is a problem with the resolver signal. The problems are typically either that the resolver is not connected or that the resolver feedback is too large or too small. All of these conditions are represented by the Resolver Not Connected Fault (Run Fault bit 30).

The resolver excitation output is a sine wave signal at a fixed frequency of 10kHz. The sine wave excitation is no longer tied to the PWM frequency. The excitation level can still be adjusted, but only to 3 different levels. The magnitude of the signal does not change with PWM frequency. The resolver excitation frequency is fixed at 10kHz.

EXC Level Setting	Nominal Excitation Voltage
5 to 7	1.95Vrms / 5.52Vpp
4	2.93Vrms / 8.28Vpp
0 to 3	3.90Vrms / 11.04Vpp

- Note level 4 is not supported currently

The SIN and COS inputs are differential inputs. The target SIN/COS input range for the Gen 4/Gen 5 products is 3.15 Vpp ± 27%. This can be measured by using an oscilloscope. Note that the signals are differential signals so it may be necessary to either use a differential scope probe or use two scope channels and subtract them. Many scopes don't allow measuring differential signals directly because the scope is grounded. It may be necessary to rotate the motor shaft to observe the maximum value.

2.9 Resolver Interface – CM Family

Important Note: The CM350SiC inverters use the RM and Gen 5 Family resolver interface. Please see section 2.8.

The CM Family resolver interface is different than the PM and RM Family.

The CM family uses a resolver to digital converter (RDC) that has more diagnostic features as well as more options for excitation level.

The resolver excitation frequency is fixed at 10kHz. The excitation frequency is not dependent on the PWM frequency.

The excitation is controlled via the Resolver_Exc_Level_EEPROM parameter. The bits of this parameter are defined in the table below. The excitation voltage level can be controlled to 9 different levels. Additionally, the SIN/COS feedback signals can have 4 different amplification levels. It is possible to have 36 different combinations. Not all the combinations are useful.

The excitation level, a 16-bit number, is set according to the table below:

Bits 15 thru 6	Bits 5 and 4			Bits 3 thru 0				
	Bit 5	Bit 4	Input Gain	Bit 3	Bit 2	Bit 1	Bit 0	Excitation Voltage (Vpp)
Always 0	0	0	0.188	0	0	0	0	8.38
	0	1	0.250	0	0	0	1	8.75
	1	0	0.563	0	0	1	0	9.11
	1	1	Don't use	0	0	1	1	9.47
				0	1	0	0	9.84
				0	1	0	1	10.20
				0	1	1	0	10.57
				0	1	1	1	10.93
				1	0	0	0	11.30

		1	0	0	1	Don't use
		1	0	1	0	Don't use
		1	0	1	1	Don't use
		1	1	0	0	Don't use
		1	1	0	1	Don't use
		1	1	1	0	Don't use
		1	1	1	1	Don't use

The target for this RDC is to have a signal that has a Target Signal voltage between 0.6 and 1.5Vpp.

Signal Voltage = Resolver transformation ratio * excitation voltage * input gain

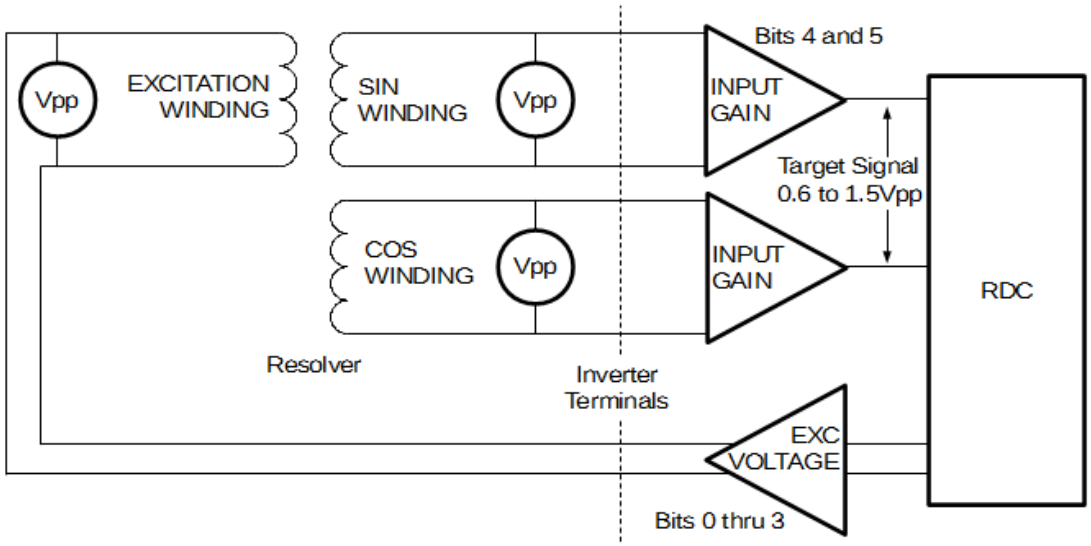


Diagram of CM Family Resolver Interface

The Target Signal voltage can be determined by measuring the resolver sin/cos winding output voltage. As noted above. Winding voltage multiplied by the Input Gain will give the Target Signal Level. The resolver winding voltages must be measured using a differential voltage measurement device (handheld meter or scope with differential capability).

The Target Signal Level is given as the level when the SIN/COS winding voltages are their maximum value. To achieve the maximum value the motor may need to be rotated to a new position.

As an example, assume a resolver transformation ratio of 0.286. With an excitation voltage of 11.30Vpp and an input gain of 0.25 the resulting signal would be 0.81Vpp (0.286 * 11.30 * 0.25). The voltage is at the lower end of acceptable, but OK. If the gain of 0.563 is used then the excitation level can be

lowered to 8.38 resulting in a signal of 1.35Vpp (0.286 * 8.38 * 0.563), at the upper end of the acceptable range.

Example	Excitation Voltage	Input Gain	Excitation Level Setting
Low End	11.30	0.250	0x18 (24 decimal)
High End	8.38	0.563	0x20 (32 decimal)

Important Note: If the resolver interface detects a fault condition it will turn off the excitation output. When the excitation output is off then no feedback will occur on the sin/cos inputs to the inverter.

2.10 Adjustable PWM Frequency

Firmware has been created that allows the PWM frequency to be changed from the standard 12 kHz to different PWM frequencies. However, this code is a non-standard release and is not included in the normal release of PM/RM code. For Gen5 and CM inverters adjustable PWM frequency is standard. If adjusting the PWM frequency is needed, then please consult with Cascadia Motion on whether this is acceptable in the application.

The current rating of the inverter is affected by the PWM frequency.

Motor losses can be greatly affected by the PWM frequency.

The section of the manual describes adjusting the PWM frequency in more detail.

The PWM frequency is adjusted by using the PWM_Frequency_EEPROM parameter that is included in the main default defsyms file. The parameter can also be accessed via CAN parameter messages address 110).

The acceptable range of PWM frequency is 6 to 16 kHz (up to 24kHz for Gen5/CM). The frequency can only be adjusted in 1kHz increments. The main motor control loop operates at the PWM frequency. The background loop operates at a fixed rate of 3ms.

Important Note: Not all PWM frequencies are compatible with all inverters.

The PM family inverters use a resolver interface that operates at the PWM frequency. Thus when the PWM frequency is changed the resolver excitation frequency is changed. Additionally due to the nature of the excitation circuit the magnitude of the resolver excitation will change with changes in PWM frequency. As noted above the excitation level can be adjusted and this adjustment can compensate for some of the changes in the PWM frequency.

The RM Family resolver interface is a fixed frequency excitation, 10 kHz, and does not change with PWM frequency. Thus the RM Family resolver does not require a change in configuration when changing the PWM frequency.

There are some inherent limitations of the internal control system that must be accounted for when changing the PWM frequency. The PWM frequency in hertz divided by the Fbase selected must not be greater than 100. For example, 12000 / 175 = 68.6, this is less than 100 so it is acceptable.

When changing the PWM frequency it may be necessary to change certain gains. The current regulator and field weakening regulator gains do not change when the PWM frequency is changed. The internal resolver observer gains are changed to account for the PWM frequency change. The CM/Gen5 firmware accounts for this automatically with certain gains. Ki gains are represented in terms of bandwidth instead (w1 = Ki / (Kp * Tpwm)).

2.11 Angle Advance Factor

Alignment of the magnetic field may need adjustments at high speeds. One common symptom that such adjustment is needed is if the motor accelerates at high speeds when there is no load and zero torque is being commanded. This can happen when there is some D-axis current commanded by the inverter and due to angle error. Some of that D-axis current actually ends up being Q-axis current which produces torque. For PM motors D-axis current is commanded when field weakening is needed; if field weakening is enabled and if D-axis current is allowed.

Angle advance factor applies an angle adjustment based on the speed. The setting is scaled in time, where a setting of 100 would be a value of one Control period. A setting of 50 would be ½ Control period. The amount of electrical angle shift in degrees can be calculated as:

$$Angle\ (°) = \frac{Angle\ Advance\ Factor}{100} * Tcontrol * 360 * \frac{Motor\ rpm * Motor\ Poles}{120}$$

The control period, Tcontrol depends on the PWM rate. Typically control rate is the same as the PWM rate. Standard PWM rate for Gen 3 inverters is 12kHz. For customer specific motors and applications may use a different PWM frequency.

A back-to-back motor dyno is needed to calibrate the angle advance factor. The test motor needs to be shaft coupled to another motor with a torque sensor in between the motors.

In theory for both PM and IPM motors, if there is zero Q-axis current and only D-axis current then there should be no torque produced. Therefore, if only D-axis current is commanded and torque is produced, then the assumption is that there exists some angle error; which is causing some of that D-axis current to actually be Q-axis current. This assumption will be used for tuning the angle advance factor. There may be other reasons for torque being produced other than angle error. For example, in some cases ripple current and variations in the current due to current regulation may also sometimes cause torque to be produced when there is zero Q-axis current commanded. However, to simplify the tuning process, those other effects will be ignored.

Before tuning angle advance factor, the resolver should already be calibrated. Start with Gamma_Adjust_EEPROM set to the value found in calibration. Also start with Angle_Advance_Factor_EEPROM set to zero. The inverter will need to be setup for direct control of Id and Iq commands over CAN. See section 2.5 Direct Control of Id and Iq Commands, above.

The test will involve applying a fixed D-axis current. Typically -100A D-axis current is used for this testing; however, if 100A is close to max or exceeds the max current of the motor then a lower current should be used.

Select about 5 different speeds to operate at. The operating speeds should be between zero and field weakening speed. Avoid operating in the field weakening region for this testing. The other coupled motor needs to be in speed mode. Use the coupled motor to spin the test motor at the desired operating points. For each of the selected operating speeds do the following steps to collect data.

1. With inverter on the test motor disabled, record the torque.
2. Enable the inverter on the test motor and command D-axis current. Zero Q-axis current.
3. Adjust gamma_adjust parameter such that the measured torque is the same as when the inverter was disabled. Record the gamma_adjust value.

Create a plot with gamma_adjust in degrees on the Y-axis and electrical speed in Hertz on the X-axis. This is the gamma_adjust found in step 3 above. To calculate electrical speed, use the following equation.

$$\omega_{elec}[Hz] = \omega_{mech}[rpm] * Poles/120$$

Where, ω_{elec} electrical speed, ω_{mech} is the mechanical speed, and poles is the total number of motor poles, or two times the number of pole pairs. Units are in brackets.

Create a linear curve fit of gamma_adjust vs electrical speed. The offset of the curve fit is the new gamma_adjust in degrees. For each operating speed calculate the angle advance factor needed using the following equation.

$$Angle_Advance_Factor_Needed = \frac{gamma_adjust[deg] - new_gamma_adjust[deg]}{angle_per_control_loop[deg]}$$

Where gamma_adjust is the gamma_adjust that results in the same torque as when inverter is disabled; as found in step 3 above. new_gamma_adjust is what was found from the offset of the curve fit done above. Angle per control loop is the electrical angle that is rotated for every control loop. This can be calculated using the following equation.

$$angle_per_contorl_loop[deg] = \frac{\omega_{elec}[Hz] * 360}{control_rate[Hz]}$$

Create a plot with angle_advance_factor_needed on the Y-axis and electrical speed in Hertz on the X-axis. Using the plot select an angle advance factor that is closest to most of the data points on the plot.

Program the new Gamma_Adjust and Angle_Advance_Factor into EEPROM. Verify that with the new settings, measured torque with D-axis current is about the same as measured torque when inverter is disabled for each of the operating speeds. Also run the motor at 1,000 RPM with inverter disabled, and record the delta_resolver_in parameter. It may no longer be 900 or 90 degrees, but should still be within +/-20 degrees. It is normal for delta_resolver_in to change due to the angle advance factor calibration. If the resolver ever needs to be calibrated again it should be calibrated to this new value instead of the 90 degrees value.

Revision History

Version	Description of Versions / Changes	Responsible Party	Date
0.1	Initial version	Chris Brune	3/24/2015
0.2	Added bit 7 to motor type option to allow setting an angle offset. Added Resolver excitation EEPROM parameter.	Chris Brune	10/20/2015
0.3	Added the Kp Current Slope and Offset EEPROM parameters.	Chris Brune	7/12/2016
0.4	Added temperature sensor type 12.	Chris Brune	7/21/2016
0.5	Add setting for direct control of id and iq current commands. Added documentation relating to adjusting the control angle.	Chris Brune	11/18/2016
0.6	Added 220K NTC Thermistor. Added to firmware in version 1993.	Chris Brune	12/13/2016
0.7	Added section to clarify the current regulator.	Chris Brune	1/4/2017
0.8	Added CAN Parameter addresses to EEPROM, valid with software version 19A4.	Chris Brune	6/14/2017
0.9	Additional clarification of Id / Iq control for IPM motors.	Chris Brune	11/7/2017
1.0	Added information that is specific to the RM Family. Clarified the resolver settings on the PM and RM Family.	Chris Brune	3/15/2018
1.1	Added Saturation Effect section	Chris Brune	7/11/2018
1.2	Updated Saturation Effect section. Added explanation of the modulation limiter	Chris Brune	7/21/2018
1.3	Add Phase Advance feature	Chris Brune	8/1/2018
1.4	Corrected error in calculation of angle advance.	Chris Brune	8/24/2018
1.5	Add section on adjusting the PWM frequency.	Chris Brune	8/28/2018
1.6	Added Cascadia Motion. Added additional details on temperature sensor types.	Chris Brune	8/12/2019
1.7	Added information on temperature sensor 25. Removed references to RMS. General edits for clarity.	Chris Brune	4/1/2020
1.8	Added temperature sensor types 27 and 28.	Chris Brune	4/23/2020

1.9	Added detail on the RM/Gen5 Resolver excitation level.	Chris Brune	5/1/2020
2.0	Added details on Angle Advance Factor feature.	Andrew Louie	10/6/2020
2.1	Added section about CM resolver interface. Added information about Gen 5 resolver interface.	Chris Brune	3/10/2021
2.2	Corrected mistake in the RM/Gen 5 resolver excitation levels.	Chris Brune	4/1/2021
2.3	Added detail about w1 versus Ki on Gen5/CM Inverters.	Chris Brune	8/18/2021
2.4	Added details on motor type options bit 11 for how to use sin/cos encoder on RM, CM, and Gen5 inverters.	Andrew Louie	8/27/2021
2.5	Fixed typo in CM Family input gain levels table. Added diagram of CM Family resolver interface.	Chris Brune	10/18/2021
2.6	Updated Motor Type Options EEPROM to include bit 15 – Swap phases. Updated Temperature Sensor Type list to include iM-225/SS-250 motor configurations.	Chris Brune	6/13/2022
2.7	Fix error in Angle Advance Factor calculation. Clarification of text.	Chris Brune	8/3/2023
2.8	Added note about CM350SiC resolver interface deviation from “CM Family”	Noah Erickson	3/28/2024